Mathematical and Computational Methods

Exercise Sheet 8: Differentiation II: applications

This sheet accompanies *Lecture 8 (Differentiation II: applications)*. It exercises the four core skills of the unit—tangents and normals, locating and classifying stationary points, curve sketching, and applied optimisation—together with the supporting theorems (intermediate-value, extreme-value, mean-value and inverse-function), l'Hôpital's rule and the idea of gradient descent. Everything here uses only the differentiation of Units 7–8 and the Unit 1 foundations; no material from the complex-number, linear-algebra or integration units is needed.

How to use this sheet. Problems are graded by difficulty—⋆ drill (routine fluency), ⋆⋆ core (standard, tutorial level) and ⋆⋆⋆ challenge (harder or extension)—and organised in three sections: **warm-up drills** for fluency, **tutorial problems** (the core set for the 2-hour class, with the live items flagged ▶) and **further practice** for independent home study.

Solutions. This is the *student* sheet (problems only); a full worked solution sits after each problem but is hidden. To produce the solutions version, uncomment `\solutionson` in the preamble (or build with `pdflatex "\def\showsolutions{}\input{MCM_Exercises_Unit8}"`) and recompile.

Contents

1 Warm-up drills

Exercise 1.1 *(⋆ drill, tangents & normals)*. For $f(x) = x^2$, find the equations of the tangent and the normal to the curve at the point $(2, 4)$.

Exercise 1.2 *(⋆ drill, l'Hôpital)*. Evaluate, stating that the form is indeterminate before applying the rule:

$$\text{(a)} \lim_{x\to 0} \frac{\sin x}{x}, \qquad \text{(b)} \lim_{x\to 0} \frac{e^x - 1}{x}, \qquad \text{(c)} \lim_{x\to\infty} \frac{\ln x}{x^2}.$$

Exercise 1.3 *(⋆ drill, classify)*. Locate and classify the stationary points of $f(x) = x^3 - 3x^2$ using the second-derivative test.

Exercise 1.4 *(⋆ drill, asymptotes)*. Find the vertical and horizontal asymptotes of

$$\text{(a)}\ f(x) = \frac{1}{x-2}, \qquad \text{(b)}\ g(x) = \frac{2x}{x-3}.$$

Exercise 1.5 *(⋆ drill, optimisation)***.** Two non-negative numbers add to 10. What is the largest their product can be?

Exercise 1.6 *(⋆ drill, gradient descent)***.** Apply gradient descent to $f(x) = x^2$ with learning rate $\eta = 0.1$, starting from $x_0 = 1$. Compute x_1 and x_2, and state the limit of the iteration.

Exercise 1.7 *(⋆ drill, critical point where f' undefined)***.** Show that $f(x) = x^{2/3}$ has a critical point at $x = 0$ even though $f'(0)$ does not exist, and decide whether it is a local maximum or minimum.

Exercise 1.8 *(⋆ drill, intermediate value theorem)***.** Show that $x^3 + x - 1 = 0$ has a solution in the interval $(0, 1)$.

Exercise 1.9 *(⋆ drill, inverse-function derivative)***.** Let $f(x) = x^5 + x + 1$. Given that f is one-to-one, find $\left(f^{-1}\right)'(1)$ *without* finding f^{-1} explicitly.

Exercise 1.10 *(⋆ drill, sketch from f')***.** Sketch the parabola $f(x) = x^2 - 4x + 3$ by first using its derivative: find the interval(s) of increase and decrease and the vertex, then add the intercepts.

Exercise 1.11 *(⋆ drill, mean value theorem)***.** Find the value of c guaranteed by the mean value theorem for $f(x) = x^2$ on $[1, 3]$.

Exercise 1.12 *(⋆ drill, extreme-value / closed interval)***.** Use the closed-interval (extreme-value-theorem) method to find the absolute maximum and minimum of $f(x) = x^2 - 4x + 1$ on $[0, 3]$.

2 Tutorial problems

The problems flagged ▶ form the ⋆⋆ core set worked live in the 2-hour tutorial; the remaining problems are for completion at home.

Exercise 2.1 *(▶ ⋆⋆ core, tangents & normals)***.** Consider the curve $y = x^3 - 2x + 1$.

1. Find the tangent and normal lines at the point where $x = 1$.
2. Find the equations of *all* tangent lines to the curve that are parallel to the line $y = x$.

Exercise 2.2 *(▶ ⋆⋆ core, l'Hôpital — several forms)***.** Evaluate, first rewriting each into a $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form where necessary:

$$\text{(a)}\ \lim_{x\to 0} \frac{x - \sin x}{x^3}, \qquad \text{(b)}\ \lim_{x\to 0^+} x \ln x, \qquad \text{(c)}\ \lim_{x\to 0} \left(\frac{1}{x} - \frac{1}{\sin x}\right).$$

Exercise 2.3 *(▶ ⋆⋆ core, classify — worked-example anchor)***.** Locate and classify all stationary points of

$$f(x) = 2x^3 - 3x^2 - 12x + 5.$$

Exercise 2.4 *(▶ ⋆⋆ core, full polynomial sketch)***.** Carry out a full curve-sketching analysis of $f(x) = x^3 - 6x^2 + 9x$ (domain/symmetry, intercepts, increase/decrease and extrema, concavity and inflection) and sketch the curve.

Exercise 2.5 *(⋆⋆ core, full rational sketch)***.** Sketch $f(x) = \dfrac{x^2 + 1}{x^2 - 1}$: state the domain, symmetry, intercepts, asymptotes, stationary points and concavity, then draw the curve.

Exercise 2.6 *(▶ ⋆⋆ core, optimisation — worked-example anchor)***.** An open-topped box is made from a square sheet of card of side 12 cm by cutting a small square of side x from each corner and folding up the flaps. Find the value of x that maximises the volume, and the maximum volume.

Exercise 2.7 *(⋆⋆ core, physics: maximum power transfer)***.** A battery of EMF ε and internal resistance r delivers power $P(R) = \dfrac{\varepsilon^2 R}{(R+r)^2}$ to an external load of resistance $R > 0$. Show that the power delivered is greatest when $R = r$, and find that maximum power.

Exercise 2.8 *(⋆⋆ core, physics: projectile range)***.** A projectile is launched from level ground with fixed speed v at an angle θ to the horizontal. Neglecting air resistance, its range is

$$R(\theta) = \frac{v^2}{g}\sin(2\theta), \qquad 0 \le \theta \le \tfrac{\pi}{2}.$$

Find the launch angle that maximises the range, and the maximum range.

Exercise 2.9 *(⋆⋆ core, data science: least squares and the mean)***.** Given data values $x_1, \dots, x_n$, the *sum of squared errors* of a single predicted constant c is $L(c) = \sum_{i=1}^{n}(x_i - c)^2$.

1. Show that L is minimised when c equals the mean $\bar{x} = \frac{1}{n}\sum_i x_i$.
2. Find the minimising c for the data $\{2, 4, 9\}$.

Exercise 2.10 *(▶ ⋆⋆ core, gradient descent on a quadratic)***.** Minimise $f(x) = x^2 - 4x + 5$ by gradient descent with learning rate $\eta = 0.25$, starting from $x_0 = 0$.

1. Write the update rule $x_{n+1} = g(x_n)$ explicitly and compute x_1, x_2, x_3.
2. To what value do the iterates converge, and why is that the right answer?
3. What would go wrong if the learning rate were taken as $\eta = 1.5$?

Exercise 2.11 *(⋆⋆ core, mean value theorem)***.**

1. Find the value of c guaranteed by the mean value theorem for $f(x) = x^3$ on $[0, 2]$.
2. Use the mean value theorem to prove that $|\sin a - \sin b| \le |a - b|$ for all real a, b.

Exercise 2.12 *(⋆⋆ core, inverse-function theorem)***.** Let $f(x) = 2x + \cos x$.

1. Show that f is strictly increasing on $\mathbb{R}$, hence one-to-one and invertible.
2. Compute $(f^{-1})'(1)$.

Exercise 2.13 *(▶ ⋆⋆ core, spot the error (conceptual))***.** To find the largest area of a rectangle with perimeter 20, a student sets width x, height $10 - x$, writes $A(x) = x(10 - x)$, finds $A'(x) = 10 - 2x = 0$, so $x = 5$, and then concludes "$A''(x) = -2 < 0$, so this is a minimum." Identify the error and state the correct conclusion.

Exercise 2.14 *(⋆⋆ core, Newton's method)***.** The equation $x^3 - 2x - 5 = 0$ has a single real root, which the intermediate value theorem traps in $(2, 3)$ (since $f(2) = -1 < 0$ and $f(3) = 16 > 0$). Use *Newton's method* to pin it down: starting from $x_0 = 2$, write the iteration explicitly and carry out two steps by hand, tabulating x_n, $f(x_n)$ and $f'(x_n)$ at each stage to obtain x_1 and x_2. Comment on how fast the estimate is settling.

3 Further practice (home)

Tangents and normals

Exercise 3.1 *(⋆⋆ core, tangent & normal)*. For the curve $y = \dfrac{1}{x}$, find the tangent and normal at the point $(2, \frac{1}{2})$, and find where the normal crosses the x-axis.

Exercise 3.2 *(⋆⋆ core, tangents from an external point)*. Find the equations of the two tangent lines to the parabola $y = x^2$ that pass through the point $(0, -1)$.

Exercise 3.3 *(⋆⋆⋆ challenge, challenge: a chord property)*. Show that the tangent to $y = x^3$ at the point (a, a^3) (with $a \neq 0$) meets the curve again at exactly one further point, and that its x-coordinate is $-2a$.

Stationary points and classification

Exercise 3.4 *(⋆⋆ core, classify a mixed bag)*. Locate and classify all stationary points of:

$$\text{(a) } f(x) = x^3 - 3x^2 + 3x, \qquad \text{(b) } g(x) = x + \frac{1}{x}, \qquad \text{(c) } h(x) = x\mathrm{e}^{-x}.$$

Exercise 3.5 *(⋆ drill, when the second-derivative test fails)*. For $f(x) = x^4$, show that $f''(0) = 0$, so the second-derivative test is inconclusive, and then classify the stationary point at $x = 0$ by another method.

Exercise 3.6 *(⋆⋆ core, absolute extrema on a closed interval)*. Use the closed-interval (extreme-value-theorem) method to find the absolute maximum and minimum of:

$$\text{(a) } f(x) = x^3 - 3x \text{ on } [0, 3], \qquad \text{(b) } g(x) = x^4 - 8x^2 + 3 \text{ on } [-1, 3].$$

Curve sketching

Exercise 3.7 *(⋆⋆ core, polynomial sketch)*. Sketch $f(x) = x^3 - 12x$: give symmetry, intercepts, stationary points and their type, and the concavity.

Exercise 3.8 *(⋆⋆ core, rational sketch)*. Sketch $f(x) = \dfrac{x}{x^2 - 1}$: domain, symmetry, intercepts, asymptotes, monotonicity.

Exercise 3.9 *(⋆⋆ core, rational sketch)*. Sketch $f(x) = \dfrac{x - 1}{x + 2}$, identifying the asymptotes and the intercepts.

Exercise 3.10 *(⋆⋆ core, sketch from features)*. Sketch $f(x) = x^4 - 8x^2 + 16$. (Hint: it factorises.)

Optimisation

Exercise 3.11 *(⋆⋆ core, inscribed rectangle)*. A rectangle has its base on the x-axis and its two upper corners on the parabola $y = 12 - x^2$ (with $y \geq 0$). Find the dimensions of the rectangle of greatest area, and that area.

Exercise 3.12 *(⋆⋆ core, divided pen)*. A rectangular animal pen of area $600\,\mathrm{m}^2$ is divided into two equal halves by a fence parallel to one pair of sides. Find the dimensions that minimise the total length of fencing used (perimeter plus the internal divider).

Exercise 3.13 *(⋆⋆ core, cylindrical can)*. A closed cylindrical can must hold a volume of $16\pi\,\mathrm{cm}^3$. Find the radius and height that minimise its total surface area, and verify that at the optimum the height equals the diameter.

Exercise 3.14 *(⋆⋆ core, data science: closest point)*. Find the point(s) on the parabola $y = x^2$ closest to $\left(0, \frac{3}{2}\right)$.

Exercise 3.15 *(⋆⋆ core, physics: kinematics min/max)*. A particle moves along a straight line with position $s(t) = t^3 - 9t^2 + 24t$ metres at time t seconds, for $0 \le t \le 5$. Find (a) the times at which it is instantaneously at rest, and (b) its maximum and minimum velocity on the interval.

Exercise 3.16 *(⋆⋆⋆ challenge, challenge: the Norman window)*. A "Norman" window is a rectangle topped by a semicircle, the semicircle's diameter being the top of the rectangle. For a fixed perimeter P, find the proportions that maximise the area, and show that at the optimum the rectangle's height equals the radius of the semicircle.

Limits (l'Hôpital)

Exercise 3.17 *(⋆⋆ core, a batch of limits)*. Evaluate:

$$\text{(a)}\ \lim_{x\to 0}\frac{\tan x - x}{x^3}, \quad \text{(b)}\ \lim_{x\to 1}\frac{\ln x}{x-1}, \quad \text{(c)}\ \lim_{x\to\infty}\frac{x^2}{\mathrm{e}^x}, \quad \text{(d)}\ \lim_{x\to 0^+} x^x.$$

Exercise 3.18 *(⋆⋆ core, spot the error)*. A student writes

$$\lim_{x\to 0}\frac{x^2+1}{x+1} \overset{\mathrm{H}}{=} \lim_{x\to 0}\frac{2x}{1} = 0.$$

Explain what is wrong, and give the correct value.

Exercise 3.19 *(⋆⋆⋆ challenge, challenge: a 1^∞ limit)*. Show that $\lim_{x\to\infty}\left(1+\frac{a}{x}\right)^x = \mathrm{e}^a$ for a constant a.

Theorems: IVT, MVT, inverse functions

Exercise 3.20 *(⋆⋆ core, IVT)*. Show that the equation $\cos x = x$ has at least one solution in $(0, 1)$.

Exercise 3.21 *(⋆⋆ core, existence and uniqueness)*. Show that $x^3 + x - 1 = 0$ has *exactly* one real root. (Hint: IVT for existence, the sign of the derivative for uniqueness.)

Exercise 3.22 *(⋆⋆ core, MVT inequality)*. Use the mean value theorem (or the sign of a derivative) to prove that $\mathrm{e}^x \ge 1 + x$ for all x, with equality only at $x = 0$.

Exercise 3.23 *(⋆⋆ core, inverse-function derivatives)*.

1. For $f(x) = x^3 + 2x + 1$ (one-to-one), find $\left(f^{-1}\right)'(1)$.
2. Use the inverse-function theorem to recover $\frac{\mathrm{d}}{\mathrm{d}x}\arctan x = \dfrac{1}{1+x^2}$ from tan.

Gradient descent

Exercise 3.24 *(⋆⋆ core, descent with a shifted minimum)*. Apply gradient descent to $f(x) = (x-3)^2$ with $\eta = 0.1$, starting from $x_0 = 0$. Write the update rule, compute x_1, x_2, x_3, and identify the limit.

Exercise 3.25 *(⋆⋆⋆ challenge, challenge: convergence condition)*. For $f(x) = x^2$ the gradient-descent update with learning rate η is $x_{n+1} = x_n - \eta\,(2x_n)$.

1. Show that $x_n = (1 - 2\eta)^n x_0$.
2. Deduce the exact range of $\eta > 0$ for which $x_n \to 0$ for every starting point, and describe the behaviour when $\eta = 1$ and when $\eta > 1$.

Conceptual / true–false

Exercise 3.26 *(⋆ drill, true or false)*. For each statement, say whether it is true or false and justify briefly.

1. If $f'(c) = 0$ then f has a local maximum or minimum at c.
2. If $f''(c) = 0$ then c is an inflection point of f.
3. A continuous function on an open interval need not attain a maximum value.
4. If $f'(x) > 0$ on an interval, then f is strictly increasing there.
5. At a point where $f'(a) = 0$, the normal line is horizontal.
6. l'Hôpital's rule can be applied to any limit of a quotient.

Exercise 3.27 *(⋆⋆ core, interpret a graph of f')*. The graph below shows the *derivative* f' of a function f (not f itself). Using only this graph, describe f: state the intervals on which f is increasing or decreasing, locate and classify its stationary points, and say where f is concave up or down and where it has a point of inflection.